Pearson Edexcel A-Level Mathematics (9MA0)

Module 4

Calculus & Vectors

Paper 1 & 2 (Pure) · 108 practice questions · answers and mark schemes at the back

Subtopic	Questions	Done
4.1 Differentiation	36	____ / 36
4.2 Integration	36	____ / 36
4.3 Numerical Methods	18	____ / 18
4.4 Vectors (3D)	18	____ / 18

How to use this workbook

1. Read the key facts and watch one video for the subtopic. Tick the skills checklist as you go.
2. Attempt each question in the squared working space without notes. Mark it with the scheme at the back.
3. Tick **Todo** when you have attempted it and **Mastered** only when you can redo it cold a week later.
4. Copy any mistakes into the notes box at the end of the subtopic. Rebuild the workbook with a new seed for fresh questions.

Name _____ Started _____ Built 14 Oct 2026, seed 2026

4.1 Differentiation

Differentiation from first principles, standard derivatives, chain/product/quotient rules, implicit and parametric differentiation, stationary points, tangents, optimisation and rates of change.

Key facts

- $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$
- $\frac{d}{dx}x^n = nx^{n-1}$, $\frac{d}{dx}e^{kx} = ke^{kx}$, $\frac{d}{dx}\ln x = \frac{1}{x}$, $\frac{d}{dx}a^x = a^x \ln a$
- $\frac{d}{dx}\sin kx = k\cos kx$, $\frac{d}{dx}\cos kx = -k\sin kx$, $\frac{d}{dx}\tan kx = k\sec^2 kx$
- Chain: $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$; product: $u'v + uv'$; quotient: $\frac{vu' - uv'}{v^2}$
- Parametric: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$; implicit: $\frac{d}{dx}y^2 = 2y\frac{dy}{dx}$
- Concave: $f''(x) < 0$; convex: $f''(x) > 0$; point of inflection: f'' changes sign

Method

- Stationary points: solve $f'(x) = 0$, then test with $f''(x)$ (or the gradient either side).
- Tangent: $y - y_1 = f'(x_1)(x - x_1)$; normal uses $-\frac{1}{f'(x_1)}$.
- Connected rates: chain the known rate with a derivative from a formula, e.g. $\frac{dr}{dt} = \frac{dV}{dt} \div \frac{dV}{dr}$.
- Optimisation: write the quantity in one variable using the constraint, differentiate, set to 0, justify max/min.

Exam traps

- Using degrees in trig derivatives (must be radians).
- Forgetting $\frac{dy}{dx}$ on y terms in implicit differentiation.
- Not justifying the nature of a stationary point when asked.

In the exam: The largest topic on Papers 1 and 2: expect 15-25 marks per paper across several questions.

Specification checklist

Skill	Learnt	Practised	Mastered
Differentiation from first principles	☐	☐	☐
Derivatives of sin x and cos x	☐	☐	☐
Exponentials and logarithms	☐	☐	☐
Chain rule	☐	☐	☐
Product rule	☐	☐	☐
Quotient rule	☐	☐	☐
Parametric differentiation	☐	☐	☐
Implicit differentiation	☐	☐	☐
Second derivatives, stationary points, convex / concave, inflection	☐	☐	☐
Tangents and normals	☐	☐	☐
Optimisation	☐	☐	☐
Connected rates of change	☐	☐	☐

Videos (click to open)

☐ **Recap** — Differentiation (Year 1) in less than 30 minutes • A-Level Maths, Pure Year 1, Chapter 12 (Bicen Maths)
☐ **Recap** — Differentiation (Year 2) in less than 42 minutes • A-Level Maths, Pure Year 2, Chapter 9 (Bicen Maths)
☐ Differentiation from first principles — Differentiation: Differentiation From First Principles (TLMaths)
☐ Derivatives of sin x and cos x — Differentiation: Differentiate sin(x) from First Principles (TLMaths)
☐ Exponentials and logarithms — Differentiation: Differentiating ln(kx) (TLMaths)
☐ Chain rule — Differentiation: Introducing the Chain Rule (TLMaths)
☐ Product rule — Differentiation: Introducing the Product Rule (TLMaths)
☐ Quotient rule — Differentiation: Introducing the Quotient Rule (TLMaths)
☐ Parametric differentiation — Differentiation: Introducing Parametric Differentiation (TLMaths)
☐ Implicit differentiation — Differentiation: Introducing Implicit Differentiation (TLMaths)
☐ Second derivatives, stationary points, convex / concave, inflection — Gradients: Finding where a Curve is Convex & Concave (TLMaths)

☐ Tangents and normals — Gradients: Introducing Tangents and Normals (TLMaths)
☐ Optimisation — Optimisation: Rectangular Fence Problem (TLMaths)
☐ Connected rates of change — Differentiation: Connected Rates of Change Example 1 (TLMaths)

Q4.1.01 Exponentials and logarithms ●○○ [3 marks] ☐ Todo ☐ Mastered

Given $y = 8^x + 4e^{3x} + \ln(4x)$, find $\frac{dy}{dx}$.

(3)

WORKING

Q4.1.02 Quotient rule ●○○ [3 marks] ☐ Todo ☐ Mastered

Use the quotient rule to find $\frac{dy}{dx}$ when $y = \frac{4x+2}{x+5}$, simplifying the numerator.

(3)

WORKING

Q4.1.03 Product rule ●○○ [3 marks] ☐ Todo ☐ Mastered

Find $\frac{dy}{dx}$ when $y = (x-4)^2\cos 2x$.

(3)

WORKING

Q4.1.04 Tangents and normals ●○○ [4 marks] ☐ Todo ☐ Mastered

Find an equation of the tangent to the curve $y = 2x^2 + 3x - 1$ at the point where $x = 4$.

(4)

WORKING

Q4.1.05 Chain rule ●○○ [2 marks] ☐ Todo ☐ Mastered

Differentiate $y = \sqrt{x^3 - 3}$ with respect to x.

(2)

WORKING

Q4.1.06 Exponentials and logarithms ●○○ [3 marks] ☐ Todo ☐ Mastered

Given $y = 5^x + e^{2x} - 4\ln(4x)$, find $\frac{dy}{dx}$.

(3)

WORKING

Q4.1.07 Quotient rule ●○○ [3 marks] ☐ Todo ☐ Mastered

Use the quotient rule to find $\frac{dy}{dx}$ when $y = \frac{x^2 + 3}{e^x + 1}$, simplifying the numerator.

(3)

WORKING

Q4.1.08 Product rule ●○○ [3 marks] ☐ Todo ☐ Mastered

Find $\frac{dy}{dx}$ when $y = (x + 1)^2 \sin 2x$.

(3)

WORKING

Q4.1.09 Tangents and normals ●○○ [4 marks] ☐ Todo ☐ Mastered

Find an equation of the normal to the curve $y = 2\sqrt{x} + 2x^2 - x - 5$ at the point where $x = 4$.

(4)

WORKING

Q4.1.10 Chain rule ●○○ [2 marks] ☐ Todo ☐ Mastered

Differentiate $y = \ln(2x - 2)$ with respect to x.

(2)

WORKING

Q4.1.11 Exponentials and logarithms ●○○ [3 marks] ☐ Todo ☐ Mastered

Given $y = 3e^{4x} - 3\ln(4x)$, find $\frac{dy}{dx}$.

(3)

WORKING

Q4.1.12 Quotient rule ●○○ [3 marks] ☐ Todo ☐ Mastered

Use the quotient rule to find $\frac{dy}{dx}$ when $y = \frac{4x+2}{e^x+1}$, simplifying the numerator.

(3)

WORKING

Q4.1.13 **Product rule** ●○○ **[3 marks]** ☐ **Todo** ☐ **Mastered**

Find $\frac{dy}{dx}$ when $y = x^3 \cos 2x$.

(3)

WORKING

Q4.1.14 **Tangents and normals** ●○○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Find an equation of the tangent to the curve $y = 3x^2 - 5x + 1$ at the point where $x = 4$.

(4)

WORKING

Q4.1.15 **Chain rule** ●○○ **[2 marks]** ☐ **Todo** ☐ **Mastered**

Differentiate $y = (3x - 4)^6$ with respect to x.

(2)

WORKING

Q4.1.16 Second derivatives, stationary points, convex / concave, inflection ●●○ [8 marks] ☐ Todo ☐ Mastered

The curve C has equation $y = -x^3 + 6x^2 - 9x - 6$.

(a) Find the coordinates of the stationary points of C and determine their nature. **(6)**

(b) Find the range of values of x for which C is concave. **(2)**

WORKING

Q4.1.17 Connected rates of change ●●○ [4 marks] ☐ Todo ☐ Mastered

Air is pumped into a spherical balloon at $12\ \text{cm}^3\ \text{s}^{-1}$. Find the rate at which the radius is increasing when the radius is $10\ \text{cm}$.

(4)

WORKING

Q4.1.18 Parametric differentiation ●●○ [4 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = t^3$, $y = t^3 - t$.
Find $\frac{dy}{dx}$ in terms of t, and its value when $t = -1$.

(4)

WORKING

Q4.1.19 **Implicit differentiation** ●●○ **[5 marks]** ☐ Todo ☐ Mastered

A curve has equation $x^2 - 3xy + 2y^2 - 3y = 5$.

(a) Find $\frac{dy}{dx}$ in terms of x and y. **(4)**

(b) Find the gradient of the curve at the point $(-3, -2)$. **(1)**

WORKING

Q4.1.20 **Differentiation from first principles** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Prove, from first principles, that the derivative of $x^2 - 6x - 1$ is $2x - 6$.

(4)

WORKING

Q4.1.21 **Second derivatives, stationary points, convex / concave, inflection** ●●○ **[8 marks]** ☐ Todo ☐ Mastered

The curve C has equation $y = -x^3 - 3x^2 + 24x - 6$.

(a) Find the coordinates of the stationary points of C and determine their nature. **(6)**

(b) Find the range of values of x for which C is concave. **(2)**

WORKING

Q4.1.22 Connected rates of change ●●○ [4 marks] ☐ Todo ☐ Mastered

Air is pumped into a spherical balloon at $2\text{ cm}^3\text{ s}^{-1}$. Find the rate at which the radius is increasing when the radius is 8 cm.

(4)

WORKING

Q4.1.23 Parametric differentiation ●●○ [4 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 2\sin t$, $y = 4t - 1$.
Find $\frac{dy}{dx}$ in terms of t, and its value when $t = \frac{\pi}{3}$.

(4)

WORKING

Q4.1.24 Implicit differentiation ●●○ [7 marks] ☐ Todo ☐ Mastered

A curve has equation $x^2 - 2xy + y^2 - 2y = 6$.

(a) Find $\frac{dy}{dx}$ in terms of x and y. **(4)**

(b) Find the gradient of the curve at the point $(-3, -1)$. **(1)**

(c) Find the equation of the normal at this point in the form $y = mx + c$. **(2)**

WORKING

Q4.1.25 Differentiation from first principles ●●○ [5 marks] ☐ Todo ☐ Mastered

Prove, from first principles, that the derivative of $-3x^3$ is $-9x^2$.

(5)

WORKING

Q4.1.26 Second derivatives, stationary points, convex / concave, inflection ●●○ [8 marks] ☐ Todo ☐ Mastered

The curve C has equation $y = x^3 - \frac{15x^2}{2} + 12x + 2$.

(a) Find the coordinates of the stationary points of C and determine their nature. **(6)**

(b) Find the range of values of x for which C is concave. **(2)**

WORKING

Q4.1.27 Connected rates of change ●●○ [3 marks] ☐ Todo ☐ Mastered

A circular ripple spreads so that its radius increases at 0.3 m s^{-1}. Find the rate of increase of its area when the radius is 4 m.

(3)

WORKING

Q4.1.28 Parametric differentiation ●●○ [4 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = e^t + 2$, $y = 4t + 2$.
Find $\frac{dy}{dx}$ in terms of t, and its value when $t = 1$.

(4)

WORKING

Q4.1.29 Implicit differentiation ●●○ [7 marks] ☐ Todo ☐ Mastered

A curve has equation $x^2 + xy + 3y^2 - 2y = 37$.

(a) Find $\frac{dy}{dx}$ in terms of x and y. **(4)**

(b) Find the gradient of the curve at the point $(-1, -3)$. **(1)**

(c) Find the equation of the normal at this point in the form $y = mx + c$. **(2)**

WORKING

Q4.1.30 Differentiation from first principles ●●○ [4 marks] ☐ Todo ☐ Mastered

Prove, from first principles, that the derivative of $-2x^2 + 5x - 1$ is $5 - 4x$.

(4)

WORKING

Q4.1.31 Second derivatives, stationary points, convex / concave, inflection ●●○ [8 marks] ☐ Todo ☐ Mastered

The curve C has equation $y = x^3 + \frac{3x^2}{2} - 6x - 3$.

(a) Find the coordinates of the stationary points of C and determine their nature. **(6)**

(b) Find the range of values of x for which C is concave. **(2)**

WORKING

Q4.1.32 Connected rates of change ●●○ [4 marks] ☐ Todo ☐ Mastered

Air is pumped into a spherical balloon at $18\,\text{cm}^3\,\text{s}^{-1}$. Find the rate at which the radius is increasing when the radius is 7 cm.

(4)

WORKING

Q4.1.33 Derivatives of sin x and cos x ●●● [5 marks] ☐ Todo ☐ Mastered

Given that θ is in radians, prove, from first principles, that the derivative of $\cos 3x$ is $-3\sin 3x$.
You may assume that as $h \to 0$, $\frac{\sin h}{h} \to 1$ and $\frac{\cos h - 1}{h} \to 0$.

(5)

WORKING

Q4.1.34 Optimisation ●●● [9 marks] ☐ Todo ☐ Mastered

A closed cylindrical can has volume 2000 cm^3 and radius r cm.

(a) Show that the surface area is $A = 2\pi r^2 + \frac{4000}{r}$. **(3)**

(b) Find the value of r that minimises A and the minimum area, to 3 significant figures. **(4)**

(c) Justify that this is a minimum. **(2)**

WORKING

Q4.1.35 Optimisation ●●● [7 marks] ☐ Todo ☐ Mastered

An open box is made from a square sheet of card of side 12 cm by cutting a square of side x cm from each corner and folding up the sides.

(a) Show that the volume is $V = x(12 - 2x)^2$. **(2)**

(b) Find the value of x that maximises V and the maximum volume. **(5)**

WORKING

Q4.1.36 Derivatives of sin x and cos x ●●● [5 marks] ☐ Todo ☐ Mastered

Given that θ is in radians, prove, from first principles, that the derivative of $\sin x$ is $\cos x$.
You may assume that as $h \to 0$, $\frac{\sin h}{h} \to 1$ and $\frac{\cos h - 1}{h} \to 0$.

(5)

WORKING

MY NOTES — 4.1 DIFFERENTIATION: MISTAKES TO REMEMBER

4.2 Integration

Integration: standard integrals, $f(ax+b)$, reverse chain rule, trig identities, substitution, parts, partial fractions, areas, the trapezium rule and differential equations.

Key facts

- $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c \ (n \neq -1)$, $\int \frac{1}{x}\,dx = \ln|x| + c$
- $\int e^{kx}\,dx = \frac{1}{k}e^{kx} + c$, $\int \cos kx\,dx = \frac{1}{k}\sin kx + c$, $\int \sin kx\,dx = -\frac{1}{k}\cos kx + c$
- $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$
- By parts: $\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$
- Trapezium rule: $\int_a^b y\,dx \approx \frac{h}{2}[y_0 + y_n + 2(y_1 + \ldots + y_{n-1})]$
- Separable: $\frac{dy}{dx} = f(x)g(y) \Rightarrow \int \frac{1}{g(y)}\,dy = \int f(x)\,dx$

Method

- Choose u in parts as the thing that gets simpler when differentiated (LATE: Logs, Algebra, Trig, Exponentials); with $\ln x$, always $u = \ln x$.
- Substitution: change dx and the limits; never mix x and u.
- Areas below the x-axis come out negative; split the integral at the roots.
- Area between curves: $\int(\text{top} - \text{bottom})\,dx$.

Exam traps

- Forgetting $+c$ (or the modulus in $\ln|x|$).
- Not changing limits in a substitution.
- Over/underestimate: trapezium rule overestimates for convex curves.

In the exam: Expect at least one 8-12 mark integration question per pure paper; DE modelling questions are common in Paper 2.

Specification checklist

Skill	Learnt	Practised	Mastered
Standard integrals	☐	☐	☐
Integrating f(ax + b)	☐	☐	☐
Reverse chain rule	☐	☐	☐
Using trig identities in integration	☐	☐	☐
Integration by substitution	☐	☐	☐
Integration by parts	☐	☐	☐
Integration with partial fractions	☐	☐	☐
Areas under curves and between curves	☐	☐	☐
Integration as the limit of a sum	☐	☐	☐
Trapezium rule	☐	☐	☐
Separable differential equations	☐	☐	☐
Modelling with differential equations	☐	☐	☐

Videos (click to open)

☐ **Recap** — Integration (Year 1) in less than 25 minutes • A-Level Maths, Pure Year 1, Chapter 13 (Bicen Maths)
☐ Standard integrals — Integration: Introducing the Fundamental Theorem of Calculus (TLMaths)
☐ Integrating f(ax + b) — Integration : $(ax+b)^n$ type functions (ExamSolutions)
☐ Reverse chain rule — Integration : $(ax+b)^n$ type functions (ExamSolutions)
☐ Using trig identities in integration — Integration using trigonometric identities (1) (ExamSolutions)
☐ Integration by substitution — How to Integrate by substitution (1) (ExamSolutions)
☐ Integration by parts — Integration by Parts - example 1 (ExamSolutions)
☐ Integration with partial fractions — Integration of Partial Fractions: Example 1 (TLMaths)
☐ Areas under curves and between curves — Integration: Finding the Area between Two Curves Example 1 (TLMaths)
☐ Integration as the limit of a sum — Integration: Integration as the Limit of a Sum (TLMaths)

☐ Trapezium rule — Numerical Integration: Introducing the Trapezium Rule (TLMaths)
☐ Separable differential equations — Differential Equations: The Method of Separation of Variables (TLMaths)
☐ Modelling with differential equations — Differential Equations: A Worded Example 1 (TLMaths)

Q4.2.01 Standard integrals ●○○ [3 marks] ☐ Todo ☐ Mastered

Find $\int\left(-6x^2 - 3 + \frac{3}{x^2}\right)dx$, simplifying each term.

(3)

WORKING

Q4.2.02 Trapezium rule ●○○ [4 marks] ☐ Todo ☐ Mastered

Use the trapezium rule with 4 strips of equal width to estimate $\int_0^2 10e^{-x^2}\,dx$, giving your answer to 3 significant figures. State whether your answer is an over- or under-estimate, given the shape of the curve.

(4)

WORKING

Q4.2.03 Integration as the limit of a sum ●○○ [3 marks] ☐ Todo ☐ Mastered

Write $\lim_{\delta x \to 0} \sum_{x=0}^{4} \sqrt{x}\,\delta x$ as an integral and hence find its exact value.

(3)

WORKING

Q4.2.04 Integrating f(ax + b) ●○○ [2 marks] ☐ Todo ☐ Mastered

Find $\int (4x - 1)^5\,dx$.

(2)

WORKING

Q4.2.05 Standard integrals ●○○ [3 marks] ☐ Todo ☐ Mastered

Find $\int\left(2x^2 - 1 + \frac{4}{x^2}\right)dx$, simplifying each term.

(3)

WORKING

Q4.2.06 Trapezium rule ●○○ [4 marks] ☐ Todo ☐ Mastered

Use the trapezium rule with 5 strips of equal width to estimate $\int_0^{\frac{5}{2}} 10e^{-x^2}\,dx$, giving your answer to 3 significant figures. State whether your answer is an over- or under-estimate, given the shape of the curve.

(4)

WORKING

Q4.2.07 Integration as the limit of a sum ●○○ [3 marks] ☐ Todo ☐ Mastered

Write $\lim_{\delta x \to 0} \sum_{x=2}^{5} x^2\,\delta x$ as an integral and hence find its exact value.

(3)

WORKING

Q4.2.08 Integrating f(ax + b) ●○○ [2 marks] ☐ Todo ☐ Mastered

Find $\int (4x-7)^3\,dx$.

(2)

WORKING

Q4.2.09 Standard integrals ●○○ [4 marks] ☐ Todo ☐ Mastered

Find the exact value of $\int_2^4 \left(x^2+\frac{1}{x}-\frac{4}{x^2}\right)dx$.

(4)

WORKING

Q4.2.10 Trapezium rule ●○○ [4 marks] ☐ Todo ☐ Mastered

Use the trapezium rule with 6 strips of equal width to estimate $\int_0^3 x\sqrt{x+1}\,dx$, giving your answer to 3 significant figures. State whether your answer is an over- or under-estimate, given the shape of the curve.

(4)

WORKING

Q4.2.11 Integration as the limit of a sum ●○○ [3 marks] ☐ Todo ☐ Mastered

Write $\lim_{\delta x \to 0} \sum_{x=0}^{4} x^2\,\delta x$ as an integral and hence find its exact value.

(3)

WORKING

Q4.2.12 Integrating f(ax + b) ●○○ [2 marks] ☐ Todo ☐ Mastered

Find $\int \frac{1}{2x+5}\,dx$.

(2)

WORKING

Q4.2.13 Areas under curves and between curves ●●○ [6 marks] ☐ Todo ☐ Mastered

The line $y = 3x + 1$ and the curve $y = x^2 + 5x - 2$ intersect at two points.

(a) Find the x-coordinates of the points of intersection. **(2)**

(b) Find the exact area of the region enclosed between the line and the curve. **(4)**

WORKING

Q4.2.14 Reverse chain rule ●●○ [3 marks] ☐ Todo ☐ Mastered

Find $\int \frac{2x+1}{x^2+x+2}\,dx$.

(3)

WORKING

Q4.2.15 Modelling with differential equations ●●○ [2 marks] ☐ Todo ☐ Mastered

The rate at which the temperature θ of a liquid falls is proportional to the difference between θ and room temperature $20\,℃$. Write down a differential equation to model this situation, defining any constant you use.

(2)

WORKING

Q4.2.16 Integration with partial fractions ●●○ [5 marks] ☐ Todo ☐ Mastered

Find the exact value of $\int_0^3 \frac{4}{x^2+4x+3}\,dx$, giving your answer as a single logarithm.

(5)

WORKING

Q4.2.17 Using trig identities in integration ●●○ [3 marks] ☐ Todo ☐ Mastered

Find $\int \tan^2 x\,dx$.

(3)

WORKING

Q4.2.18 Separable differential equations ●●○ [4 marks] ☐ Todo ☐ Mastered

Find the particular solution of $\frac{dy}{dx} = \frac{1x^2}{y}$, $y > 0$, for which $y = 1$ when $x = 0$.

(4)

WORKING

Q4.2.19 Integration by parts ●●○ [4 marks] ☐ Todo ☐ Mastered

Use integration by parts to find $\int x\cos 2x\,dx$.

(4)

WORKING

Q4.2.20 Areas under curves and between curves ●●○ [6 marks] ☐ Todo ☐ Mastered

The line $y = -x - 2$ and the curve $y = 2x^2 - 13x + 8$ intersect at two points.

(a) Find the x-coordinates of the points of intersection. **(2)**

(b) Find the exact area of the region enclosed between the line and the curve. **(4)**

WORKING

Q4.2.21 Reverse chain rule ●●○ [3 marks] ☐ Todo ☐ Mastered

Find $\int x(x^2 + 4)^6\,dx$.

(3)

WORKING

Q4.2.22 **Modelling with differential equations** ●●○ **[2 marks]** ☐ **Todo** ☐ **Mastered**

The rate at which the volume V of a raindrop decreases is proportional to its surface area, which is proportional to $V^{\frac{2}{3}}$. Write down a differential equation to model this situation, defining any constant you use.

(2)

WORKING

Q4.2.23 **Integration with partial fractions** ●●○ **[5 marks]** ☐ **Todo** ☐ **Mastered**

Find the exact value of $\int_0^2 \frac{-5x-7}{x^2+3x+2}\,dx$, giving your answer as a single logarithm.

(5)

WORKING

Q4.2.24 **Using trig identities in integration** ●●○ **[3 marks]** ☐ **Todo** ☐ **Mastered**

Find $\int \tan^2 3x\,dx$.

(3)

WORKING

Q4.2.25 Separable differential equations ●●○ [5 marks] ☐ Todo ☐ Mastered

A cup of tea cools in a room at $21\,℃$. Its temperature θ after t minutes satisfies $\frac{d\theta}{dt} = -0.07(\theta - 21)$. Initially $\theta = 71$.

(a) Show that $\theta = 21 + 50e^{-0.07t}$. **(4)**

(b) Find the temperature after 12 minutes. **(1)**

WORKING

Q4.2.26 Integration by parts ●●○ [4 marks] ☐ Todo ☐ Mastered

Use integration by parts to find $\int xe^{3x}\,dx$.

(4)

WORKING

Q4.2.27 Areas under curves and between curves ●●○ [6 marks] ☐ Todo ☐ Mastered

The line $y = 2x - 1$ and the curve $y = x^2 - 5x + 9$ intersect at two points.

(a) Find the x-coordinates of the points of intersection. **(2)**

(b) Find the exact area of the region enclosed between the line and the curve. **(4)**

WORKING

Q4.2.28 Reverse chain rule ●●○ [3 marks] ☐ Todo ☐ Mastered

Find $\int x(x^2+3)^3\,dx$.

(3)

WORKING

Q4.2.29 Integration with partial fractions ●●○ [5 marks] ☐ Todo ☐ Mastered

Find the exact value of $\int_0^3 \frac{-5x-14}{x^2+6x+8}\,dx$, giving your answer as a single logarithm.

(5)

WORKING

Q4.2.30 Using trig identities in integration ●●○ [3 marks] ☐ Todo ☐ Mastered

Find $\int \tan^2 4x\,dx$.

(3)

WORKING

Q4.2.31 Separable differential equations ●●○ [5 marks] ☐ Todo ☐ Mastered

Solve $\frac{dy}{dx} = 2xy$, given $y = 3$ when $x = 0$. Give y in terms of x.

(5)

WORKING

Q4.2.32 Integration by parts ●●○ [4 marks] ☐ Todo ☐ Mastered

Use integration by parts to find $\int x \sin 3x \, dx$.

(4)

WORKING

Q4.2.33 Areas under curves and between curves ●●○ [6 marks] ☐ Todo ☐ Mastered

The line $y = 2x - 2$ and the curve $y = x^2 + 5x - 2$ intersect at two points.

(a) Find the x-coordinates of the points of intersection. **(2)**

(b) Find the exact area of the region enclosed between the line and the curve. **(4)**

WORKING

Q4.2.34 Integration by substitution ●●● [4 marks] ☐ Todo ☐ Mastered

Use the substitution $u = x^2 + 2$ to find $\int \frac{x}{(x^2+2)^2} \, dx$.

(4)

WORKING

Q4.2.35 Integration by substitution ●●● [6 marks] ☐ Todo ☐ Mastered

Use the substitution $u = x + 4$ to find the exact value of $\int_{-3}^{8} x\sqrt{x+4}\, dx$.

(6)

WORKING

Q4.2.36 Integration by substitution ●●● [6 marks] ☐ Todo ☐ Mastered

Use the substitution $u = x + 5$ to find the exact value of $\int_{-5}^{7} x\sqrt{x+5}\, dx$.

(6)

WORKING

MY NOTES — 4.2 INTEGRATION: MISTAKES TO REMEMBER

4.3 Numerical Methods

Numerical methods: locating roots by change of sign, fixed-point iteration (and its staircase/cobweb diagrams), and the Newton-Raphson method.

Key facts

- If f is continuous on $[a, b]$ and $f(a)f(b) < 0$, there is a root in (a, b)
- Iteration $x_{n+1} = g(x_n)$ converges near the root if $|g'(\alpha)| < 1$
- Staircase when $0 < g'(\alpha) < 1$; cobweb when $-1 < g'(\alpha) < 0$
- Newton-Raphson: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

Method

- To show a root is 2.345 to 3 d.p., check the sign of $f(2.3445)$ and $f(2.3455)$.
- Always state 'sign change and f is continuous, so a root lies in the interval'.
- Newton-Raphson fails when $f'(x_n) = 0$ (horizontal tangent) or the start is near a turning point.

Exam traps

- Forgetting the continuity statement.
- Rounding iterates too early.
- Using degrees in trig iterations.

In the exam: Paper 2, 5-8 marks; often the end of a calculus or modelling question.

Specification checklist

Skill	Learnt	Practised	Mastered
Locating roots by change of sign	☐	☐	☐
When change of sign fails	☐	☐	☐
Fixed-point iteration x = g(x)	☐	☐	☐
Staircase and cobweb diagrams	☐	☐	☐
Newton-Raphson method	☐	☐	☐
Applications to modelling	☐	☐	☐

Videos (click to open)

☐ **Recap** — Numerical Methods in 21 minutes • A-Level Maths, Pure Year 2, Chapter 10 (Bicen Maths)
☐ Locating roots by change of sign — Locating Roots: Introducing the Change of Sign Method (TLMaths)
☐ When change of sign fails — Locating Roots: Failure of the Change of Sign Method (TLMaths)
☐ Fixed-point iteration x = g(x) — Locating Roots: Introducing the x=g(x) Method (TLMaths)
☐ Staircase and cobweb diagrams — Locating Roots: Cobweb and Staircase Diagrams (TLMaths)
☐ Newton-Raphson method — Locating Roots: Introducing the Newton-Raphson Method (TLMaths)
☐ Applications to modelling — Numerical Methods: Contextual Problem 2 (TLMaths)

Q4.3.01 Locating roots by change of sign ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 + 3x^2 + 5x + 1$. Show that the equation $f(x) = 0$ has a root α in the interval $[-1, 0]$.

(2)

WORKING

Q4.3.02 Locating roots by change of sign ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 + 4x^2 - 6x + 7$. Show that the equation $f(x) = 0$ has a root α in the interval $[-6, -5]$.

(2)

WORKING

Q4.3.03 Locating roots by change of sign ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 - 3x^2 + 6x + 2$. Show that the equation $f(x) = 0$ has a root α in the interval $[-1, 0]$.

(2)

WORKING

Q4.3.04 Locating roots by change of sign ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 + 2x^2 - x - 7$. Show that the equation $f(x) = 0$ has a root α in the interval $[1, 2]$.

(2)

WORKING

Q4.3.05 When change of sign fails ●●○ [2 marks] ☐ Todo ☐ Mastered

Explain why the change of sign method fails to locate the root of $f(x) = (x + 2)^2$ and why considering $f(x) = \frac{1}{x+2}$ on $[-3, -1]$ would be misleading.

(2)

WORKING

Q4.3.06 Staircase and cobweb diagrams ●●○ [2 marks] ☐ Todo ☐ Mastered

An iteration $x_{n+1} = g(x_n)$ has a fixed point α, and $g'(\alpha) = -2$. State whether the iteration converges to α for a starting value close to α, and whether the diagram is a staircase or a cobweb.

(2)

WORKING

Q4.3.07 Fixed-point iteration x = g(x) ●●○ [6 marks] ☐ Todo ☐ Mastered

(a) Show that $x^3 - 4x - 2 = 0$ can be written as $x = \sqrt[3]{4x+2}$. **(1)**

(b) Using $x_{n+1} = \sqrt[3]{4x_n + 2}$ with $x_0 = 2$, find x_1, x_2 and x_3 to 4 decimal places. **(3)**

(c) Show that the root is 2.214 to 3 decimal places. **(2)**

WORKING

Q4.3.08 Newton-Raphson method ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 + 3x^2 + 4x + 5$. Taking $x_0 = -2.5$ as a first approximation to a root of $f(x) = 0$, apply the Newton-Raphson method twice to find x_2 to 3 decimal places.

(4)

WORKING

Q4.3.09 **When change of sign fails** ●●○ **[2 marks]** ☐ Todo ☐ Mastered

Explain why the change of sign method fails to locate the root of $f(x) = (x-4)^2$ and why considering $f(x) = \frac{1}{x-4}$ on $[3, 5]$ would be misleading.

(2)

WORKING

Q4.3.10 **Staircase and cobweb diagrams** ●●○ **[2 marks]** ☐ Todo ☐ Mastered

An iteration $x_{n+1} = g(x_n)$ has a fixed point α, and $g'(\alpha) = -\frac{3}{4}$. State whether the iteration converges to α for a starting value close to α, and whether the diagram is a staircase or a cobweb.

(2)

WORKING

Q4.3.11 **Fixed-point iteration x = g(x)** ●●○ **[6 marks]** ☐ Todo ☐ Mastered

(a) Show that $x^3 - 4x - 3 = 0$ can be written as $x = \sqrt[3]{4x+3}$. **(1)**

(b) Using $x_{n+1} = \sqrt[3]{4x_n + 3}$ with $x_0 = 2$, find x_1, x_2 and x_3 to 4 decimal places. **(3)**

(c) Show that the root is 2.303 to 3 decimal places. **(2)**

WORKING

Q4.3.12 Newton-Raphson method ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 + 4x^2 - 4x - 9$. Taking $x_0 = -4.5$ as a first approximation to a root of $f(x) = 0$, apply the Newton-Raphson method twice to find x_2 to 3 decimal places.

(4)

WORKING

Q4.3.13 When change of sign fails ●●○ [2 marks] ☐ Todo ☐ Mastered

Explain why the change of sign method fails to locate the root of $f(x) = (x+4)^2$ and why considering $f(x) = \frac{1}{x+4}$ on $[-5, -3]$ would be misleading.

(2)

WORKING

Q4.3.14 Fixed-point iteration x = g(x) ●●○ [6 marks] ☐ Todo ☐ Mastered

(a) Show that $x^3 - 5x - 7 = 0$ can be written as $x = \sqrt[3]{5x+7}$. **(1)**

(b) Using $x_{n+1} = \sqrt[3]{5x_n + 7}$ with $x_0 = 3$, find x_1, x_2 and x_3 to 4 decimal places. **(3)**

(c) Show that the root is 2.747 to 3 decimal places. **(2)**

WORKING

Q4.3.15 Newton-Raphson method ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 - x^2 + 5x + 2$. Taking $x_0 = -0.5$ as a first approximation to a root of $f(x) = 0$, apply the Newton-Raphson method twice to find x_2 to 3 decimal places.

(4)

WORKING

Q4.3.16 Applications to modelling ●●● [4 marks] ☐ Todo ☐ Mastered

The height difference between two tanks is modelled by $f(t) = e^{0.5t} - 59 + 2t$, $t \geq 0$. The tanks are level when $f(t) = 0$.
Use the Newton-Raphson method with $t_0 = 6$ to find this time to 2 decimal places.

(4)

WORKING

Q4.3.17 Applications to modelling ●●● [4 marks] ☐ Todo ☐ Mastered

The height difference between two tanks is modelled by $f(t) = e^{0.5t} - 35 + 2t$, $t \geq 0$. The tanks are level when $f(t) = 0$.
Use the Newton-Raphson method with $t_0 = 6$ to find this time to 2 decimal places.

(4)

WORKING

Q4.3.18 **Applications to modelling** ●●● **[4 marks]** ☐ **Todo** ☐ **Mastered**

The height difference between two tanks is modelled by $f(t) = e^{0.5t} - 41 + 2t$, $t \geq 0$. The tanks are level when $f(t) = 0$.

Use the Newton-Raphson method with $t_0 = 6$ to find this time to 2 decimal places.

(4)

WORKING

MY NOTES — 4.3 NUMERICAL METHODS: MISTAKES TO REMEMBER

4.4 Vectors (3D)

Vectors in 2D and 3D: magnitude and direction, position vectors, distances, geometric problems (parallel, collinear, ratios) and vectors in mechanics.

Key facts

- $|a\mathbf{i} + b\mathbf{j} + c\mathbf{k}| = \sqrt{a^2 + b^2 + c^2}$; unit vector $\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$
- $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$; midpoint $\frac{1}{2}(\mathbf{a} + \mathbf{b})$
- Angle with the x-axis: $\cos\theta_x = \frac{a}{|\mathbf{v}|}$ (similarly for y, z)
- Parallel vectors are scalar multiples; collinear points share a point and a parallel direction

Method

- For a parallelogram $ABCD$: $\overrightarrow{AD} = \overrightarrow{BC}$.
- For ratio problems, write a path in two ways and compare coefficients of non-parallel vectors.
- In mechanics, resultant force $= m\mathbf{a}$; speed is the magnitude of velocity.

Exam traps

- $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$, not $\mathbf{a} - \mathbf{b}$.
- Confusing position vector with displacement.

In the exam: Short 4-6 mark questions in pure; vector kinematics and forces in Paper 3 mechanics.

Specification checklist

Skill	Learnt	Practised	Mastered
3D coordinates and i, j, k notation	☐	☐	☐
Magnitude and direction	☐	☐	☐
Position vectors and distance	☐	☐	☐
Angles with axes	☐	☐	☐
Geometric problems with vectors	☐	☐	☐
Vectors in mechanics	☐	☐	☐

Videos (click to open)

☐ **Recap** — Vectors (Year 1) in less than 21 minutes • A-Level Maths, Pure Year 1, Chapter 11 (Bicen Maths)
☐ **Recap** — Vectors (Year 2) in less than 14 minutes • A-Level Maths, Pure Year 2, Chapter 12 (Bicen Maths)
☐ 3D coordinates and i, j, k notation — Vectors: Introducing 3D Vectors (TLMaths)
☐ Magnitude and direction — Vectors: Finding the Magnitude / Length of a 3D vector (TLMaths)
☐ Position vectors and distance — Vectors: Introducing Position Vectors (TLMaths)
☐ Angles with axes — Angle between a 3d vector and the positive axis (ExamSolutions)
☐ Geometric problems with vectors — Vectors: Vectors Problem 1 (TLMaths)
☐ Vectors in mechanics — Vectors: Kinematics Problem (TLMaths)

Q4.4.01 Position vectors and distance ●○○ [4 marks] ☐ Todo ☐ Mastered

Points A and B have position vectors $-\mathbf{j}-\mathbf{k}$ and $3\mathbf{i}+\mathbf{j}$.

(a) Find $\overrightarrow{AB}$. **(1)**

(b) Find the exact distance AB. **(2)**

(c) Find the position vector of the midpoint of AB. **(1)**

WORKING

Q4.4.02 3D coordinates and i, j, k notation ●○○ [3 marks] ☐ Todo ☐ Mastered

Show that the points $A(3, 1, 1)$, $B(6, 4, -1)$ and $C(12, 10, -5)$ are collinear, and find the ratio $AB : BC$.

(3)

WORKING

Q4.4.03 Magnitude and direction ●○○ [2 marks] ☐ Todo ☐ Mastered

$\mathbf{a} = -\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$. Find $|\mathbf{a}|$ and a unit vector in the direction of $\mathbf{a}$.

(2)

WORKING

Q4.4.04 Position vectors and distance ●○○ [4 marks] ☐ Todo ☐ Mastered

Points A and B have position vectors $3\mathbf{i}+6\mathbf{j}-6\mathbf{k}$ and $2\mathbf{i}+5\mathbf{j}+4\mathbf{k}$.

(a) Find $\overrightarrow{AB}$. **(1)**

(b) Find the exact distance AB. **(2)**

(c) Find the position vector of the midpoint of AB. **(1)**

WORKING

Q4.4.05 3D coordinates and i, j, k notation ●○○ [3 marks] ☐ Todo ☐ Mastered

Show that the points $A(4, 5, 5)$, $B(1, 7, 2)$ and $C(-8, 13, -7)$ are collinear, and find the ratio $AB : BC$.

(3)

WORKING

Q4.4.06 Magnitude and direction ●○○ [2 marks] ☐ Todo ☐ Mastered

$\mathbf{a} = -\mathbf{i} + 4\mathbf{j} - 8\mathbf{k}$. Find $|\mathbf{a}|$ and a unit vector in the direction of $\mathbf{a}$.

(2)

WORKING

Q4.4.07 Position vectors and distance ●○○ [4 marks] ☐ Todo ☐ Mastered

Points A and B have position vectors $-2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $-6\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$.

(a) Find $\overrightarrow{AB}$. **(1)**

(b) Find the exact distance AB. **(2)**

(c) Find the position vector of the midpoint of AB. **(1)**

WORKING

Q4.4.08 3D coordinates and i, j, k notation ●○○ [3 marks] ☐ Todo ☐ Mastered

Show that the points $A(4, -5, -2)$, $B(2, -6, -3)$ and $C(0, -7, -4)$ are collinear, and find the ratio $AB : BC$.

(3)

WORKING

Q4.4.09 Magnitude and direction ●○○ [2 marks] ☐ Todo ☐ Mastered

$\mathbf{a} = 3\mathbf{i} + 4\mathbf{j} + 12\mathbf{k}$. Find $|\mathbf{a}|$ and a unit vector in the direction of $\mathbf{a}$.

(2)

WORKING

Q4.4.10 Vectors in mechanics ●●○ [3 marks] ☐ Todo ☐ Mastered

Two forces $\mathbf{F}_1 = (-\mathbf{i} - \mathbf{j} - 8\mathbf{k})$ N and $\mathbf{F}_2 = (-3\mathbf{i} - 5\mathbf{j} + 3\mathbf{k})$ N act on a particle of mass 2 kg. Find the magnitude of the acceleration of the particle.

(3)

WORKING

Q4.4.11 Geometric problems with vectors ●●○ [3 marks] ☐ Todo ☐ Mastered

$ABCD$ is a parallelogram with $A(-3, 4, 4)$, $B(3, 0, 3)$ and $C(1, -3, 2)$. Find the coordinates of D.

(3)

WORKING

Q4.4.12 Angles with axes ●●○ [2 marks] ☐ Todo ☐ Mastered

Find the angle that the vector $4\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ makes with the positive z-axis, to 1 decimal place.

(2)

WORKING

Q4.4.13 Vectors in mechanics ●●○ [3 marks] ☐ Todo ☐ Mastered

Two forces $\mathbf{F}_1 = (5\mathbf{i} - 4\mathbf{j} + 8\mathbf{k})$ N and $\mathbf{F}_2 = (3\mathbf{i} - 3\mathbf{j} - 7\mathbf{k})$ N act on a particle of mass 4 kg. Find the magnitude of the acceleration of the particle.

(3)

WORKING

Q4.4.14 Geometric problems with vectors ●●○ [3 marks] ☐ Todo ☐ Mastered

$ABCD$ is a parallelogram with $A(5, -2, -5)$, $B(2, -1, -3)$ and $C(-5, -5, 2)$. Find the coordinates of D.

(3)

WORKING

Q4.4.15 Angles with axes ●●○ [2 marks] ☐ Todo ☐ Mastered

Find the angle that the vector $2\mathbf{i} - 5\mathbf{j} - 3\mathbf{k}$ makes with the positive y-axis, to 1 decimal place.

(2)

WORKING

Q4.4.16 Vectors in mechanics ●●○ [3 marks] ☐ Todo ☐ Mastered

Two forces $\mathbf{F}_1 = (6\mathbf{i} - \mathbf{j} + 6\mathbf{k})$ N and $\mathbf{F}_2 = (3\mathbf{i} - \mathbf{j} - 8\mathbf{k})$ N act on a particle of mass 3 kg. Find the magnitude of the acceleration of the particle.

(3)

WORKING

Q4.4.17 Geometric problems with vectors ●●○ [3 marks] ☐ Todo ☐ Mastered

$ABCD$ is a parallelogram with $A(4, 4, -4)$, $B(-3, -5, 4)$ and $C(-1, 4, -4)$. Find the coordinates of D.

(3)

WORKING

Q4.4.18 Angles with axes ●●○ [2 marks] ☐ Todo ☐ Mastered

Find the angle that the vector $-4\mathbf{i} + 4\mathbf{j} - 6\mathbf{k}$ makes with the positive y-axis, to 1 decimal place.

(2)

WORKING

MY NOTES — 4.4 VECTORS (3D): MISTAKES TO REMEMBER

Answers and mark schemes

M = method mark, A = accuracy mark (depends on the preceding M), B = independent mark, ft = follow-through, * = given answer. Numerical answers are to 3 s.f. unless stated.

4.1 Differentiation

Q4.1.01 **Exponentials and logarithms · 3 marks**

$\frac{dy}{dx} = 8^x \ln 8 + 12e^{3x} + \frac{1}{x}$

B1 exponential term · B1 log term · B1 remaining term

Q4.1.02 **Quotient rule · 3 marks**

$\frac{18}{(x+5)^2}$

M1 $\frac{vu' - uv'}{v^2}$ · A1 correct unsimplified · A1 simplified

Q4.1.03 **Product rule · 3 marks**

$-2(x-4)^2 \sin 2x + (2x-8)\cos 2x$

M1 $u'v + uv'$ form · A1 one term · A1 fully correct

Q4.1.04 **Tangents and normals · 4 marks**

$y = 19x - 33$

M1 differentiates · A1 gradient 19 · M1 tangent and point · A1

Q4.1.05 **Chain rule · 2 marks**

$\frac{3x^2}{2\sqrt{x^3-3}}$

M1 chain rule form · A1

Q4.1.06 **Exponentials and logarithms · 3 marks**

$\frac{dy}{dx} = 5^x \ln 5 + 2e^{2x} - \frac{4}{x}$

B1 exponential term · B1 log term · B1 remaining term

Q4.1.07 **Quotient rule · 3 marks**

$\frac{-x^2e^x + 2xe^x + 2x - 3e^x}{(e^x+1)^2}$

M1 $\frac{vu' - uv'}{v^2}$ · A1 correct unsimplified · A1 simplified

Q4.1.08 **Product rule · 3 marks**

$2(x+1)^2 \cos 2x + (2x+2)\sin 2x$

M1 $u'v + uv'$ form · A1 one term · A1 fully correct

Q4.1.09 **Tangents and normals · 4 marks**

$y = \frac{845}{31} - \frac{2x}{31}$

M1 differentiates · A1 gradient $\frac{31}{2}$ · M1 uses $-1/m$ and point · A1

Q4.1.10 **Chain rule · 2 marks**

$\frac{2}{2x-2}$

M1 chain rule form · A1

Q4.1.11 **Exponentials and logarithms · 3 marks**

$\frac{dy}{dx} = 12e^{4x} - \frac{3}{x}$

B1 exponential term · B1 log term · B1 remaining term

Q4.1.12 **Quotient rule · 3 marks**

$\frac{-4xe^x + 2e^x + 4}{(e^x+1)^2}$

M1 $\frac{vu' - uv'}{v^2}$ · A1 correct unsimplified · A1 simplified

Q4.1.13 **Product rule · 3 marks**

$-2x^3 \sin 2x + 3x^2 \cos 2x$

M1 $u'v + uv'$ form · A1 one term · A1 fully correct

Q4.1.14 **Tangents and normals · 4 marks**

$y = 19x - 47$

M1 differentiates · A1 gradient 19 · M1 tangent and point · A1

Q4.1.15 **Chain rule · 2 marks**

$18(3x-4)^5$

M1 chain rule form · A1

Q4.1.16 **Second derivatives, stationary points, convex / concave, inflection · 8 marks**

(a) $(1,\ -10)$ minimum; $(3,\ -6)$ maximum

M1 differentiates · M1 sets $\frac{dy}{dx}=0$ and solves · A1 x values · A1 y values · M1 second derivative test · A1 natures

(b) $x>2$

M1 $\frac{d^2y}{dx^2}<0$ · A1

Q4.1.17 **Connected rates of change · 4 marks**

0.00955 cm s^{-1}

B1 $\frac{dV}{dr}=4\pi r^2$ · M1 chain rule $\frac{dr}{dt}=\frac{dV}{dt}\div\frac{dV}{dr}$ · M1 substitutes · A1

Q4.1.18 **Parametric differentiation · 4 marks**

$\frac{dy}{dx}=1-\frac{1}{3t^2}$; at $t=-1$: $\frac{2}{3}$

B1 $\frac{dx}{dt}$ · B1 $\frac{dy}{dt}$ · M1 divides correctly · A1 value

Q4.1.19 **Implicit differentiation · 5 marks**

(a) $\frac{dy}{dx}=\frac{2x-3y}{3x-4y+3}$

M1 differentiates y terms with $\frac{dy}{dx}$ · B1 product rule on xy · M1 collects $\frac{dy}{dx}$ · A1

(b) 0

B1ft

Q4.1.20 **Differentiation from first principles · 4 marks**

$\lim_{h\to0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to0}(2x-6+h)=2x-6$

B1 $f(x+h)$ expanded correctly · M1 $\frac{f(x+h)-f(x)}{h}$ simplified · M1 limit as $h\to0$ taken with correct notation · A1* complete proof

Q4.1.21 **Second derivatives, stationary points, convex / concave, inflection · 8 marks**

(a) $(-4,\ -86)$ minimum; $(2,\ 22)$ maximum

M1 differentiates · M1 sets $\frac{dy}{dx}=0$ and solves · A1 x values · A1 y values · M1 second derivative test · A1 natures

(b) $x>-1$

M1 $\frac{d^2y}{dx^2}<0$ · A1

Q4.1.22 **Connected rates of change · 4 marks**

0.00249 cm s^{-1}

B1 $\frac{dV}{dr}=4\pi r^2$ · M1 chain rule $\frac{dr}{dt}=\frac{dV}{dt}\div\frac{dV}{dr}$ · M1 substitutes · A1

Q4.1.23 **Parametric differentiation · 4 marks**

$\frac{dy}{dx}=\frac{2}{\cos t}$; at $t=\frac{\pi}{3}$: 4

B1 $\frac{dx}{dt}$ · B1 $\frac{dy}{dt}$ · M1 divides correctly · A1 value

Q4.1.24 **Implicit differentiation · 7 marks**

(a) $\frac{dy}{dx}=\frac{x-y}{x-y+1}$

M1 differentiates y terms with $\frac{dy}{dx}$ · B1 product rule on xy · M1 collects $\frac{dy}{dx}$ · A1

(b) 2

B1ft

(c) $y=-\frac{x}{2}-\frac{5}{2}$

M1 normal gradient and point · A1

Q4.1.25 **Differentiation from first principles · 5 marks**

$\frac{-3(3x^2h+3xh^2+h^3)}{h}\to-9x^2$ as $h\to0$

B1 $(x+h)^3$ expanded · M1 difference quotient · M1 divides by h · M1 limit · A1*

Q4.1.26 **Second derivatives, stationary points, convex / concave, inflection · 8 marks**

(a) $(1,\ \frac{15}{2})$ maximum; $(4,\ -6)$ minimum

M1 differentiates · M1 sets $\frac{dy}{dx}=0$ and solves · A1 x values · A1 y values · M1 second derivative test · A1 natures

(b) $x<\frac{5}{2}$

M1 $\frac{d^2y}{dx^2}<0$ · A1

Q4.1.27 **Connected rates of change · 3 marks**

$7.54\text{ m}^2\text{ s}^{-1}$

B1 $\frac{dA}{dr}=2\pi r$ · M1 · A1

Q4.1.28 **Parametric differentiation · 4 marks**

$\frac{dy}{dx}=4e^{-t}$; at $t=1$: $\frac{4}{e}$

B1 $\frac{dx}{dt}$ · B1 $\frac{dy}{dt}$ · M1 divides correctly · A1 value

Q4.1.29 **Implicit differentiation · 7 marks**

(a) $\frac{dy}{dx}=\frac{-2x-y}{x+6y-2}$

M1 differentiates y terms with $\frac{dy}{dx}$ · B1 product rule on xy · M1 collects $\frac{dy}{dx}$ · A1

(b) $-\frac{5}{21}$

B1ft

(c) $y=\frac{21x}{5}+\frac{6}{5}$

M1 normal gradient and point · A1

Q4.1.30 Differentiation from first principles · 4 marks

$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}(5-4x-2h) = 5-4x$

B1 $f(x+h)$ expanded correctly · M1 $\frac{f(x+h)-f(x)}{h}$ simplified · M1 limit as $h\to 0$ taken with correct notation · A1* complete proof

Q4.1.31 Second derivatives, stationary points, convex / concave, inflection · 8 marks

(a) $(-2, 7)$ maximum; $(1, -\frac{13}{2})$ minimum

M1 differentiates · M1 sets $\frac{dy}{dx}=0$ and solves · A1 x values · A1 y values · M1 second derivative test · A1 natures

(b) $x < -\frac{1}{2}$

M1 $\frac{d^2y}{dx^2} < 0$ · A1

Q4.1.32 Connected rates of change · 4 marks

0.0292 cm s^{-1}

B1 $\frac{dV}{dr} = 4\pi r^2$ · M1 chain rule $\frac{dr}{dt} = \frac{dV}{dt} \div \frac{dV}{dr}$ · M1 substitutes · A1

Q4.1.33 Derivatives of sin x and cos x · 5 marks

uses the addition formula, splits into the two standard limits

B1 uses $\cos(A+B)$ expansion · M1 difference quotient · M1 groups terms with $\sin 3h$ and $\cos 3h - 1$ · M1 applies the limits · A1* complete

Q4.1.34 Optimisation · 9 marks

(a) $h = \frac{2000}{\pi r^2}$ substituted into $2\pi r^2 + 2\pi rh$

B1 volume equation · M1 substitutes h · A1*

(b) $r = 6.83$, $A = 879\text{ cm}^2$

M1 differentiates · M1 sets $=0$ · A1 r · A1 A

(c) $\frac{d^2A}{dr^2} = 4\pi + \frac{8000}{r^3} > 0$

M1 · A1

Q4.1.35 Optimisation · 7 marks

(a) base side $12-2x$, height x

M1 · A1*

(b) $x = 2$, $V = 128\text{ cm}^3$

M1 expands and differentiates · A1 · M1 solves $V' = 0$ (rejects $x=6$) · A1 x · A1 V

Q4.1.36 Derivatives of sin x and cos x · 5 marks

uses the addition formula, splits into the two standard limits

B1 uses $\sin(A+B)$ expansion · M1 difference quotient · M1 groups terms with $\sin h$ and $\cos h - 1$ · M1 applies the limits · A1* complete

4.2 Integration

Q4.2.01 Standard integrals · 3 marks

$-2x^3 - 3x - \frac{3}{x} + c$

M1 raises a power by 1 · A1 two terms · A1 all terms and $+c$

Q4.2.02 Trapezium rule · 4 marks

$x=0$: $y=10.0000$, $x=\frac{1}{2}$: $y=7.7880$, $x=1$: $y=3.6788$, $x=\frac{3}{2}$: $y=1.0540$, $x=2$: $y=0.1832$; estimate 8.81 (underestimate; true value 8.821)

B1 $h=\frac{1}{2}$ · B1 y values · M1 $\frac{h}{2}[y_0+y_n+2(\ldots)]$ · A1

Q4.2.03 Integration as the limit of a sum · 3 marks

$\int_0^4 \sqrt{x}\,dx = \frac{16}{3}$

B1 integral form · M1 integrates · A1

Q4.2.04 Integrating f(ax + b) · 2 marks

$\frac{(4x-1)^6}{24} + c$

M1 correct form, divides by 4 · A1

Q4.2.05 Standard integrals · 3 marks

$\frac{2x^3}{3} - x - \frac{4}{x} + c$

M1 raises a power by 1 · A1 two terms · A1 all terms and $+c$

Q4.2.06 Trapezium rule · 4 marks

$x=0$: $y=10.0000$, $x=\frac{1}{2}$: $y=7.7880$, $x=1$: $y=3.6788$, $x=\frac{3}{2}$: $y=1.0540$, $x=2$: $y=0.1832$, $x=\frac{5}{2}$: $y=0.0193$; estimate 8.86 (underestimate; true value 8.859)

B1 $h=\frac{1}{2}$ · B1 y values · M1 $\frac{h}{2}[y_0+y_n+2(\ldots)]$ · A1

Q4.2.07 Integration as the limit of a sum · 3 marks

$\int_2^5 x^2\,dx = 39$

B1 integral form · M1 integrates · A1

Q4.2.08 **Integrating f(ax + b) · 2 marks**

$\frac{(4x-7)^4}{16}+c$

M1 correct form, divides by 4 · A1

Q4.2.09 **Standard integrals · 4 marks**

$\ln 2+\frac{53}{3}$

M1 integrates · A1 · M1 substitutes limits · A1 exact

Q4.2.10 **Trapezium rule · 4 marks**

$x=0$: $y=0.0000$, $x=\frac{1}{2}$: $y=0.6124$, $x=1$: $y=1.4142$, $x=\frac{3}{2}$: $y=2.3717$, $x=2$: $y=3.4641$, $x=\frac{5}{2}$: $y=4.6771$, $x=3$: $y=6.0000$; estimate 7.77 (overestimate; true value 7.733)

B1 $h=\frac{1}{2}$ · B1 y values · M1 $\frac{h}{2}[y_0+y_n+2(\ldots)]$ · A1

Q4.2.11 **Integration as the limit of a sum · 3 marks**

$\int_0^4 x^2\,dx=\frac{64}{3}$

B1 integral form · M1 integrates · A1

Q4.2.12 **Integrating f(ax + b) · 2 marks**

$\frac{\ln|2x+5|}{2}+c$

M1 correct form, divides by 2 · A1

Q4.2.13 **Areas under curves and between curves · 6 marks**

(a) $x=-3$, $x=1$

M1 equates · A1

(b) $\frac{32}{3}$

M1 $\int(\text{line}-\text{curve})\,dx$ · M1 integrates · M1 limits · A1

Q4.2.14 **Reverse chain rule · 3 marks**

$\ln(x^2+x+2)+c$

M1 recognises $f'(x)g(f(x))$ form · M1 adjusts constant · A1

Q4.2.15 **Modelling with differential equations · 2 marks**

$\frac{d\theta}{dt}=-k(\theta-20)$, $k>0$ a constant

M1 correct derivative and proportionality · A1 sign and constant defined

Q4.2.16 **Integration with partial fractions · 5 marks**

$\ln 4$

M1 partial fractions · A1 · M1 integrates to $\ln$ terms · M1 limits · A1

Q4.2.17 **Using trig identities in integration · 3 marks**

$-x+\tan x+c$

M1 uses $\tan^2 A=\sec^2 A-1$ · M1 integrates · A1

Q4.2.18 **Separable differential equations · 4 marks**

$y=\sqrt{\frac{2x^3}{3}+1}$

M1 $\int y\,dy=\int 1x^2\,dx$ · A1 $\frac{y^2}{2}=\frac{1x^3}{3}+c$ · M1 condition · A1

Q4.2.19 **Integration by parts · 4 marks**

$\frac{x\sin 2x}{2}+\frac{\cos 2x}{4}+c$

M1 $u=x$ · A1 correct first stage · M1 integrates second term · A1

Q4.2.20 **Areas under curves and between curves · 6 marks**

(a) $x=1$, $x=5$

M1 equates · A1

(b) $\frac{64}{3}$

M1 $\int(\text{line}-\text{curve})\,dx$ · M1 integrates · M1 limits · A1

Q4.2.21 **Reverse chain rule · 3 marks**

$\frac{(x^2+4)^7}{14}+c$

M1 recognises $f'(x)g(f(x))$ form · M1 adjusts constant · A1

Q4.2.22 **Modelling with differential equations · 2 marks**

$\frac{dV}{dt}=-kV^{\frac{2}{3}}$, $k>0$ a constant

M1 correct derivative and proportionality · A1 sign and constant defined

Q4.2.23 **Integration with partial fractions · 5 marks**

$-\ln 72$

M1 partial fractions · A1 · M1 integrates to $\ln$ terms · M1 limits · A1

Q4.2.24 **Using trig identities in integration · 3 marks**

$-x+\frac{\tan 3x}{3}+c$

M1 uses $\tan^2 A=\sec^2 A-1$ · M1 integrates · A1

Q4.2.25 **Separable differential equations · 5 marks**

(a) separate, integrate to $\ln(\theta - 21) = -0.07t + c$, use $t = 0$

M1 separates · A1 · M1 condition · A1*

(b) 42.6 ℃

B1

Q4.2.26 **Integration by parts · 4 marks**

$\frac{(3x-1)e^{3x}}{9} + c$

M1 $u = x$ · A1 correct first stage · M1 integrates second term · A1

Q4.2.27 **Areas under curves and between curves · 6 marks**

(a) $x = 2$, $x = 5$

M1 equates · A1

(b) $\frac{9}{2}$

M1 $\int(\text{line} - \text{curve})\,dx$ · M1 integrates · M1 limits · A1

Q4.2.28 **Reverse chain rule · 3 marks**

$\frac{(x^2+3)^4}{8} + c$

M1 recognises $f'(x)g(f(x))$ form · M1 adjusts constant · A1

Q4.2.29 **Integration with partial fractions · 5 marks**

$\ln\left(\frac{256}{8575}\right)$

M1 partial fractions · A1 · M1 integrates to ln terms · M1 limits · A1

Q4.2.30 **Using trig identities in integration · 3 marks**

$-x + \frac{\tan 4x}{4} + c$

M1 uses $\tan^2 A = \sec^2 A - 1$ · M1 integrates · A1

Q4.2.31 **Separable differential equations · 5 marks**

$y = 3e^{x^2}$

M1 separates $\int \frac{1}{y}\,dy = \int 2x\,dx$ · A1 $\ln y = x^2 + c$ · M1 uses condition · M1 exponentiates · A1

Q4.2.32 **Integration by parts · 4 marks**

$-\frac{x\cos 3x}{3} + \frac{\sin 3x}{9} + c$

M1 $u = x$ · A1 correct first stage · M1 integrates second term · A1

Q4.2.33 **Areas under curves and between curves · 6 marks**

(a) $x = -3$, $x = 0$

M1 equates · A1

(b) $\frac{9}{2}$

M1 $\int(\text{line} - \text{curve})\,dx$ · M1 integrates · M1 limits · A1

Q4.2.34 **Integration by substitution · 4 marks**

$-\frac{1}{2x^2+4} + c$

B1 $du = 2x\,dx$ · M1 $\frac{1}{2}\int u^{-2}\,du$ · M1 integrates · A1 in terms of x

Q4.2.35 **Integration by substitution · 6 marks**

$\frac{34}{15} + \frac{256\sqrt{3}}{5}$

B1 $dx = du$, new limits · M1 $\int(u-4)u^{\frac{1}{2}}\,du$ · M1 integrates · A1 · M1 limits · A1 exact

Q4.2.36 **Integration by substitution · 6 marks**

$\frac{176\sqrt{3}}{5}$

B1 $dx = du$, new limits · M1 $\int(u-5)u^{\frac{1}{2}}\,du$ · M1 integrates · A1 · M1 limits · A1 exact

4.3 Numerical Methods

Q4.3.01 **Locating roots by change of sign · 2 marks**

$f(-1) = -2$, $f(0) = 1$; sign change and f is continuous, so a root lies in the interval

M1 evaluates both · A1 sign change, continuity, conclusion

Q4.3.02 **Locating roots by change of sign · 2 marks**

$f(-6) = -29$, $f(-5) = 12$; sign change and f is continuous, so a root lies in the interval

M1 evaluates both · A1 sign change, continuity, conclusion

Q4.3.03 **Locating roots by change of sign · 2 marks**

$f(-1) = -8$, $f(0) = 2$; sign change and f is continuous, so a root lies in the interval

M1 evaluates both · A1 sign change, continuity, conclusion

Q4.3.04 **Locating roots by change of sign · 2 marks**

$f(1) = -5$, $f(2) = 7$; sign change and f is continuous, so a root lies in the interval

M1 evaluates both · A1 sign change, continuity, conclusion

Q4.3.05 **When change of sign fails · 2 marks**
repeated root at $x = -2$: no sign change; $\frac{1}{x - -2}$ changes sign across an asymptote, not a root
B1 · B1

Q4.3.06 **Staircase and cobweb diagrams · 2 marks**
diverges ($|g'(\alpha)| > 1$); cobweb diagram
B1 convergence with reason · B1 shape

Q4.3.07 **Fixed-point iteration x = g(x) · 6 marks**
(a) rearranges
B1*
(b) $x_1 = 2.1544$, $x_2 = 2.1979$, $x_3 = 2.2098$
M1 · A1 x_1 · A1 x_2, x_3
(c) $f(2.2138)$ and $f(2.2148)$ have opposite signs
M1 interval bounds · A1 conclusion

Q4.3.08 **Newton-Raphson method · 4 marks**
$x_1 = -2.2581$, $x_2 = -2.215$
B1 $f'(x) = 3x^2 + 6x + 4$ · M1 $x_0 - \frac{f(x_0)}{f'(x_0)}$ · A1 x_1 · A1 x_2

Q4.3.09 **When change of sign fails · 2 marks**
repeated root at $x = 4$: no sign change; $\frac{1}{x - 4}$ changes sign across an asymptote, not a root
B1 · B1

Q4.3.10 **Staircase and cobweb diagrams · 2 marks**
converges ($|g'(\alpha)| < 1$); cobweb diagram
B1 convergence with reason · B1 shape

Q4.3.11 **Fixed-point iteration x = g(x) · 6 marks**
(a) rearranges
B1*
(b) $x_1 = 2.2240$, $x_2 = 2.2828$, $x_3 = 2.2977$
M1 · A1 x_1 · A1 x_2, x_3
(c) $f(2.3023)$ and $f(2.3033)$ have opposite signs
M1 interval bounds · A1 conclusion

Q4.3.12 **Newton-Raphson method · 4 marks**
$x_1 = -4.4458$, $x_2 = -4.444$
B1 $f'(x) = 3x^2 + 8x - 4$ · M1 $x_0 - \frac{f(x_0)}{f'(x_0)}$ · A1 x_1 · A1 x_2

Q4.3.13 **When change of sign fails · 2 marks**
repeated root at $x = -4$: no sign change; $\frac{1}{x - -4}$ changes sign across an asymptote, not a root
B1 · B1

Q4.3.14 **Fixed-point iteration x = g(x) · 6 marks**
(a) rearranges
B1*
(b) $x_1 = 2.8020$, $x_2 = 2.7594$, $x_3 = 2.7500$
M1 · A1 x_1 · A1 x_2, x_3
(c) $f(2.7468)$ and $f(2.7478)$ have opposite signs
M1 interval bounds · A1 conclusion

Q4.3.15 **Newton-Raphson method · 4 marks**
$x_1 = -0.3704$, $x_2 = -0.364$
B1 $f'(x) = 3x^2 - 2x + 5$ · M1 $x_0 - \frac{f(x_0)}{f'(x_0)}$ · A1 x_1 · A1 x_2

Q4.3.16 **Applications to modelling · 4 marks**
$t = 7.56$
B1 $f'(t) = 0.5e^{0.5t} + 2$ · M1 applies the formula · A1 first iterate · A1 converged value

Q4.3.17 **Applications to modelling · 4 marks**
$t = 6.23$
B1 $f'(t) = 0.5e^{0.5t} + 2$ · M1 applies the formula · A1 first iterate · A1 converged value

Q4.3.18 **Applications to modelling · 4 marks**
$t = 6.64$
B1 $f'(t) = 0.5e^{0.5t} + 2$ · M1 applies the formula · A1 first iterate · A1 converged value

4.4 Vectors (3D)

Q4.4.01 **Position vectors and distance · 4 marks**

(a) $3\mathbf{i}+2\mathbf{j}+\mathbf{k}$

B1

(b) $\sqrt{14}$

M1 · A1

(c) $\frac{1}{2}(3\mathbf{i}-\mathbf{k})$

B1

Q4.4.02 **3D coordinates and i, j, k notation · 3 marks**

$\overrightarrow{AC}=3\overrightarrow{AB}$ with common point A; $AB:BC=1:2$

M1 finds two vectors · A1 scalar multiple + common point · B1 ratio

Q4.4.03 **Magnitude and direction · 2 marks**

$|\mathbf{a}|=3$; $\frac{1}{3}(-\mathbf{i}-2\mathbf{j}+2\mathbf{k})$

B1 · B1ft

Q4.4.04 **Position vectors and distance · 4 marks**

(a) $-\mathbf{i}-\mathbf{j}+10\mathbf{k}$

B1

(b) $\sqrt{102}$

M1 · A1

(c) $\frac{1}{2}(5\mathbf{i}+11\mathbf{j}-2\mathbf{k})$

B1

Q4.4.05 **3D coordinates and i, j, k notation · 3 marks**

$\overrightarrow{AC}=4\overrightarrow{AB}$ with common point A; $AB:BC=1:3$

M1 finds two vectors · A1 scalar multiple + common point · B1 ratio

Q4.4.06 **Magnitude and direction · 2 marks**

$|\mathbf{a}|=9$; $\frac{1}{9}(-\mathbf{i}+4\mathbf{j}-8\mathbf{k})$

B1 · B1ft

Q4.4.07 **Position vectors and distance · 4 marks**

(a) $-4\mathbf{i}-3\mathbf{j}$

B1

(b) 5

M1 · A1

(c) $\frac{1}{2}(-8\mathbf{i}-5\mathbf{j}+4\mathbf{k})$

B1

Q4.4.08 **3D coordinates and i, j, k notation · 3 marks**

$\overrightarrow{AC}=2\overrightarrow{AB}$ with common point A; $AB:BC=1:1$

M1 finds two vectors · A1 scalar multiple + common point · B1 ratio

Q4.4.09 **Magnitude and direction · 2 marks**

$|\mathbf{a}|=13$; $\frac{1}{13}(3\mathbf{i}+4\mathbf{j}+12\mathbf{k})$

B1 · B1ft

Q4.4.10 **Vectors in mechanics · 3 marks**

4.39 m s^{-2}

M1 resultant force · M1 $\mathbf{F}=m\mathbf{a}$ · A1

Q4.4.11 **Geometric problems with vectors · 3 marks**

$D(-5,\,1,\,3)$

M1 $\overrightarrow{AD}=\overrightarrow{BC}$ · M1 $\overrightarrow{OD}=\overrightarrow{OA}+\overrightarrow{BC}$ · A1

Q4.4.12 **Angles with axes · 2 marks**

115.9°

M1 $\cos\theta=\frac{-2}{\sqrt{21}}$ · A1

Q4.4.13 **Vectors in mechanics · 3 marks**

2.67 m s^{-2}

M1 resultant force · M1 $\mathbf{F}=m\mathbf{a}$ · A1

Q4.4.14 **Geometric problems with vectors · 3 marks**

$D(-2,\,-6,\,0)$

M1 $\overrightarrow{AD}=\overrightarrow{BC}$ · M1 $\overrightarrow{OD}=\overrightarrow{OA}+\overrightarrow{BC}$ · A1

Q4.4.15 **Angles with axes · 2 marks**

144.2°

M1 $\cos\theta=\frac{-5}{\sqrt{38}}$ · A1

Q4.4.16 **Vectors in mechanics · 3 marks**

3.14 m s^{-2}

M1 resultant force · M1 $\mathbf{F} = m\mathbf{a}$ · A1

Q4.4.17 **Geometric problems with vectors · 3 marks**

$D(6, 13, -12)$

M1 $\overrightarrow{AD} = \overrightarrow{BC}$ · M1 $\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{BC}$ · A1

Q4.4.18 **Angles with axes · 2 marks**

61.0°

M1 $\cos\theta = \frac{4}{\sqrt{68}}$ · A1